

MUHAMMAD AHMED MUFTI

AI Engineer

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SUMMARY

AI engineer who builds and evaluates LLM systems end to end. Sole author of an empirical study of four small language models in agentic RAG (1,499 experimental cells, two 70B judges at Cohen's kappa 0.65, under USD 25 of compute), builder of production RAG and multi-agent pipelines, and the engineer behind a fine-tuned RoBERTa classifier run over 14.1 million Reddit comments. Computer Science graduate, FAST-NUCES (CGPA 3.6 / 4.00); Generative AI and Agentic AI both A+.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, C/C++
ML & Deep Learning	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas; fine-tuning, AWQ inference
LLM Systems	LangChain, LangGraph, RAG, ReAct agents, MCP tool layers, LLM-as-judge evaluation, RAGAS
Retrieval & Data	FAISS, BM25, BGE embeddings and rerankers, PostgreSQL, DuckDB
Infrastructure	Modal (serverless GPU), Groq, FastAPI, ComfyUI, Docker, Git, Linux; Wilson CIs, Cohen's kappa

RESEARCH & PROJECTS

Small Language Models in Agentic RAG for Financial Document Reasoning

2026

Independent research, sole author. Manuscript in Springer LNCS format; arXiv preprint in preparation. Code (MIT):

github.com/AhmedMufti/slm-agentic-financial-rag

- Evaluated four small language models (Qwen3-1.7B, SmolLM3-3B, Phi-4-mini, Llama-3.1-8B) and a 117B open-weight ceiling (GPT-OSS-120B) on FinanceBench under naive RAG and an agentic ReAct loop.
- Built the whole harness on Modal serverless GPUs: hybrid BM25 plus BGE-small retrieval reranked with bge-reranker-v2-m3 to the top 8 chunks, four model-serving classes, the ReAct driver, and a two-judge adjudication pipeline (Llama-3.3-70B and Qwen-2.5-72B, AWQ).
- Designed a 10-category failure-mode taxonomy; reported every accuracy with Wilson 95% confidence intervals. Found the agentic loop lowered accuracy for every model and that numerical error, not retrieval, dominated failures.

Agentic Film Studio: Multi-Agent Generative Video

2026

LangGraph, PuLID, Stable Video Diffusion, Wav2Lip, Edge-TTS, FastAPI, Modal. Code: github.com/AhmedMufti/agentic-film-studio

- Five cooperating agents (story, audio, video, edit, orchestrator) turn one prompt into a lip-synced animated short with consistent characters; every agent acts through one MCP tool layer, state is logged append-only for undo, and each stage degrades to a local fallback so a video is always produced.

MedScholar: Retrieval-Augmented Clinical Research Platform **LIVE**

2025 - 2026

LangChain, FAISS, PubMed API, FastAPI, PostgreSQL, React. Live: med-scholar-ten.vercel.app

- Answers clinical questions from PubMed literature with cited sources; hybrid FAISS retrieval with strict grounding constraints to limit hallucination. Final year project.

recs.co: Sentiment Classification at Reddit Scale **LIVE**

2025

Fine-tuned RoBERTa, Modal GPU (A100 / T4), DuckDB, FastAPI. Live: recs.co

- Fine-tuned RoBERTa and ran classification over 14.1 million comments from 937 subreddits on Modal GPU workers; rankings served through a DuckDB-backed API.

From-Scratch Deep Learning in PyTorch

2025 - 2026

No high-level wrappers: CycleGAN, Transformer, ViT, RNN seq2seq, autoencoders. Code (MIT): github.com/AhmedMufti/cyclegan-transformer-vit

- CycleGAN (9-block ResNet generator, 70x70 PatchGAN) for face-to-sketch translation with a Flask UI; English-to-Urdu Transformer (BLEU about 0.43 on 22.8k pairs) benchmarked against fine-tuned mBART-50; ViT versus CNN on CIFAR-10 (CNN 92.8%, scratch ViT 82.2%, pretrained ViT-tiny 97.2%).
- Also a vanilla RNN encoder-decoder with beam search, a denoising autoencoder on CIFAR-10 (PSNR, SSIM), and a VAE on Fashion-MNIST, written up as an LNCS-format report.

EXPERIENCE

Freelance ML / AI Engineer, Fiverr (Level 1 Seller)

Jun 2024 - Present

Independent clients, remote

- Built Brand Engine, a generative-AI service that produces brand assets on demand (Python, FastAPI, ComfyUI with Flux.1 Schnell, Modal serverless GPUs).
- Delivered machine-learning and full-stack projects for international clients and reached Level 1 Seller through consistent on-time delivery.

Teaching Assistant (Part-time)

Jan 2025 - Jun 2026

FAST-NUCES, Islamabad, Pakistan

- Ran labs, wrote assignments, and graded projects for Computer Networks, Theory of Automata, and Professional Practices in IT, supporting over 100 students; selected on academic performance.

EDUCATION

BSc Computer Science

Aug 2022 - Jun 2026

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

- CGPA 3.6 / 4.00. Dean's List for four consecutive semesters (Spring 2024 to Fall 2025). Coursework: Generative AI (A+), Agentic AI (A+), Artificial Intelligence, Statistical Modelling, Numerical Computing (A+), Probability and Statistics (A), Linear Algebra, Parallel and Distributed Computing.

ACHIEVEMENTS & CERTIFICATIONS

- Fiverr Level 1 Seller. IELTS Academic 8.0. Machine Learning Specialization (Stanford / Coursera, in progress); AI For Everyone (DeepLearning.AI).